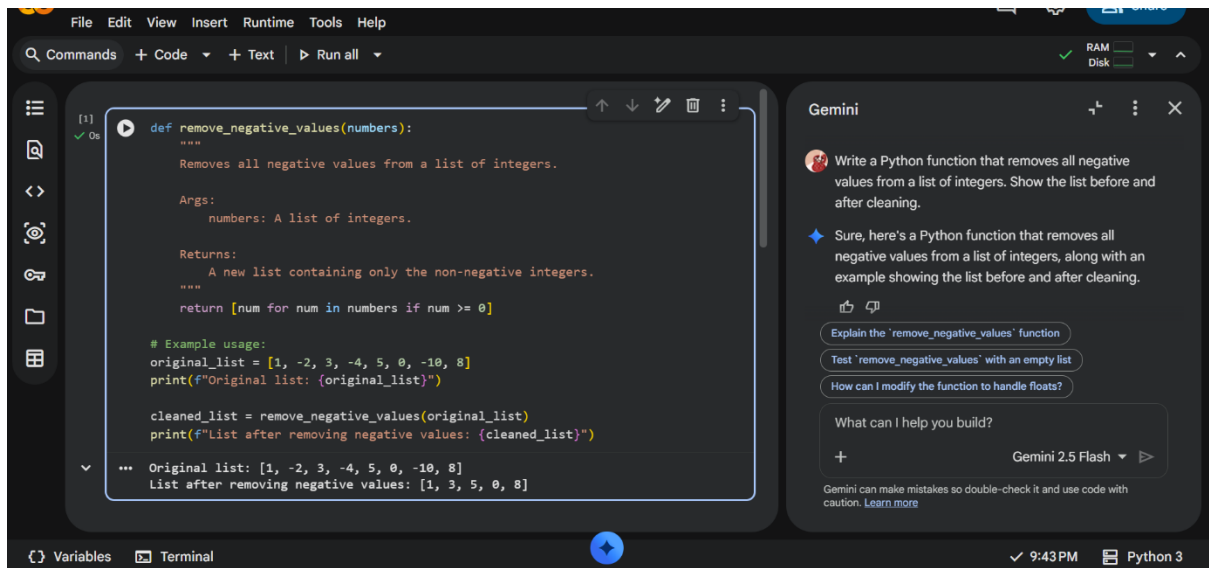


N. Shree Deepthi Batch – 37

AI-ASSISTED CODING ASSIGNMENT 2



Task 1

Prompt Used:

Write a Python function that removes all negative values from a list of integers.

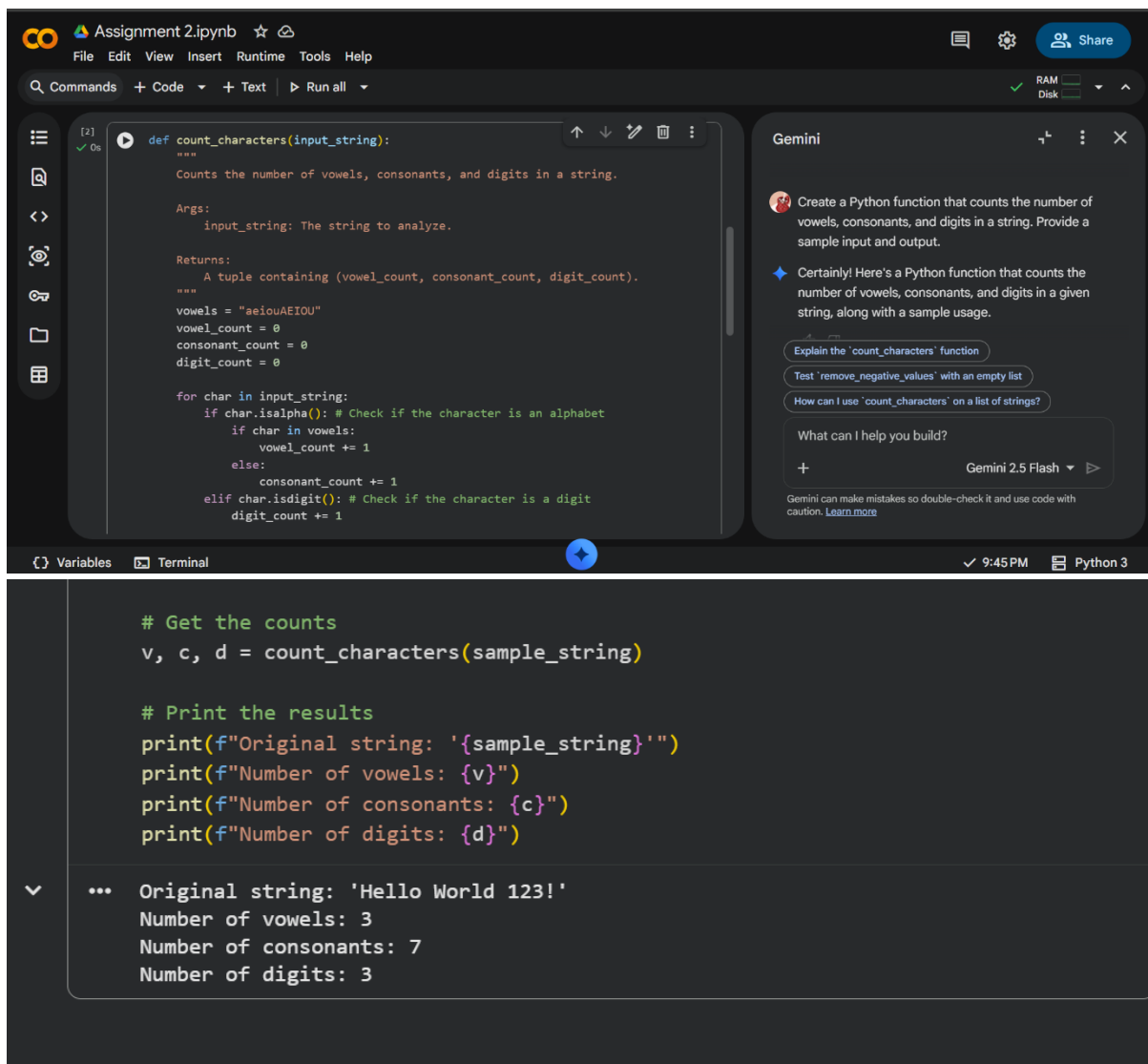
Output:

Before: [10, -3, 5, -8, 12, 0]

After: [10, 5, 12, 0]

Understanding:

Gemini generated a clean and efficient function using lists. The logic was easy to understand and it required no need for manual correction.



The screenshot shows a Jupyter Notebook interface with a dark theme. The top bar includes the file name 'Assignment 2.ipynb' and various icons for file operations, settings, and sharing. The main area is divided into two panes. The left pane contains a Python function definition for 'count_characters'. The right pane shows a Gemini chat interface with a prompt and a response. Below the chat interface, the function is called with a sample string, and the results are printed.

```
def count_characters(input_string):  
    """  
    Counts the number of vowels, consonants, and digits in a string.  
    Args:  
        input_string: The string to analyze.  
    Returns:  
        A tuple containing (vowel_count, consonant_count, digit_count).  
    """  
    vowels = "aeiouAEIOU"  
    vowel_count = 0  
    consonant_count = 0  
    digit_count = 0  
  
    for char in input_string:  
        if char.isalpha(): # Check if the character is an alphabet  
            if char in vowels:  
                vowel_count += 1  
            else:  
                consonant_count += 1  
        elif char.isdigit(): # Check if the character is a digit  
            digit_count += 1
```

Gemini

Create a Python function that counts the number of vowels, consonants, and digits in a string. Provide a sample input and output.

Certainly! Here's a Python function that counts the number of vowels, consonants, and digits in a given string, along with a sample usage.

Explain the 'count_characters' function

Test 'remove_negative_values' with an empty list

How can I use 'count_characters' on a list of strings?

What can I help you build?

+

Gemini 2.5 Flash

Gemini can make mistakes so double-check it and use code with caution. [Learn more](#)

```
# Get the counts  
v, c, d = count_characters(sample_string)  
  
# Print the results  
print(f"Original string: '{sample_string}'")  
print(f"Number of vowels: {v}")  
print(f"Number of consonants: {c}")  
print(f"Number of digits: {d}")
```

Original string: 'Hello World 123!'
Number of vowels: 3
Number of consonants: 7
Number of digits: 3

Task 2

Prompt Used :

Create a Python function that counts the number of vowels, consonants, and digits in a string.

Sample Input:

Hello123

Output:

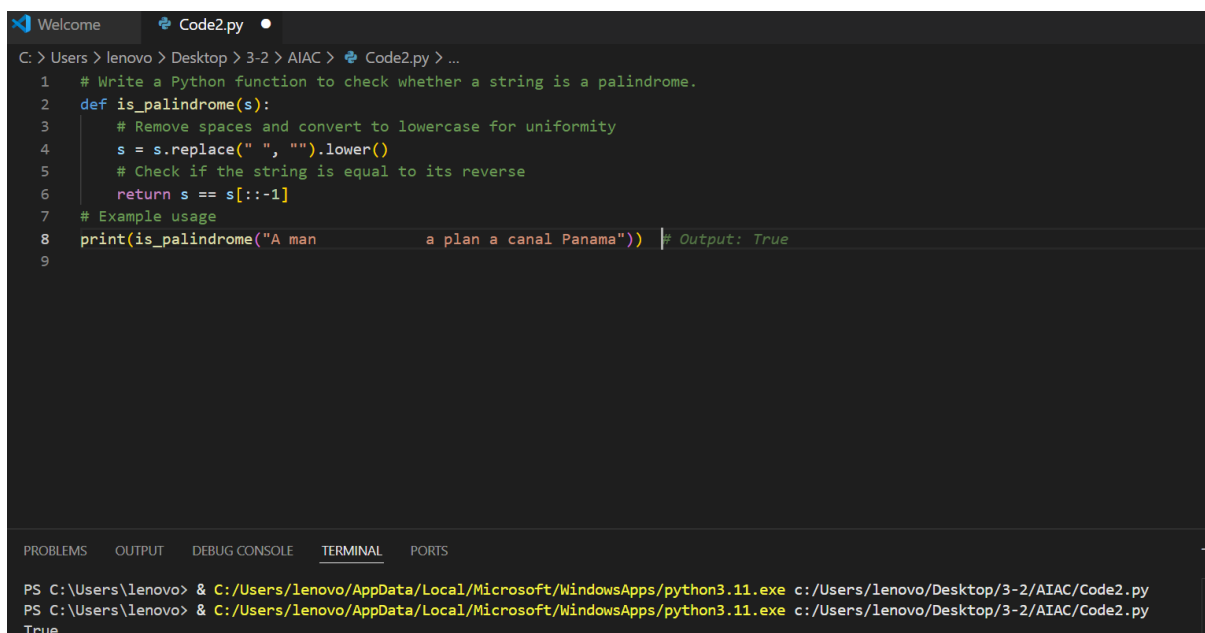
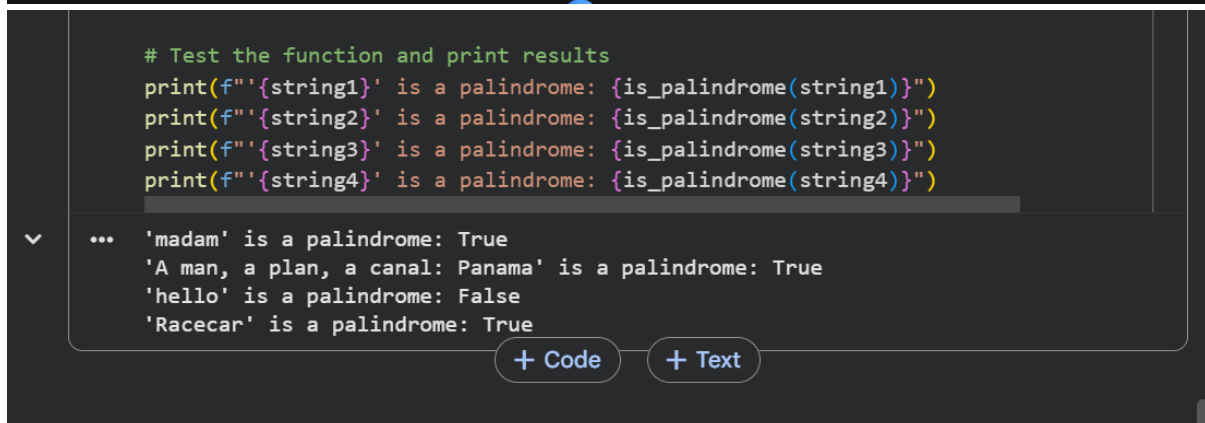
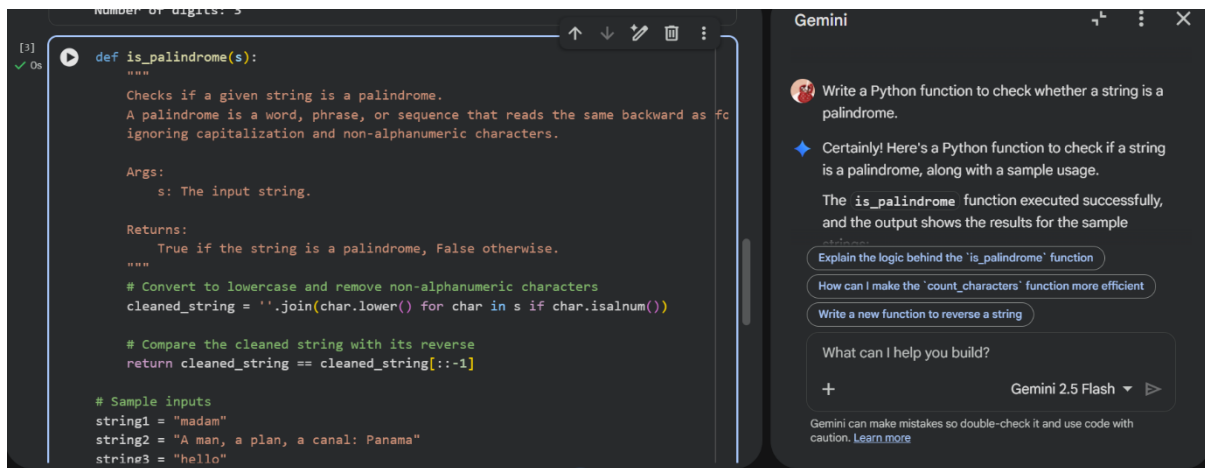
Vowels: 2

Consonants: 3

Digits: 3

Student Comment:

Gemini clearly handled character classification using built-in Python methods like `isalpha()` and `isdigit()` methods.



TASK 3:

Palindrome Check – Tool Comparison

Prompt Used:

Write a Python function to check whether a string is a palindrome.

Tool	Code Quality	Readability	Comments
Gemini	Clear and beginner-friendly	High	Explains logic step-by-step
Copilot	Short and optimized	Medium	Assumes Prior Python knowledge



Explain the following Python palindrome checking function line by line: `def is_palindrome(s): s = s.lower()
return s == s[::-1]`



Certainly! Here's a line-by-line explanation of the Python palindrome checking function:

Test the implemented `is_palindrome` function with more complex cases

Compare the explained `is_palindrome` function with the one in the notebook

Explain the `count_characters` function in the second code cell

What can I help you build?



Gemini 2.5 Flash  

Gemini can make mistakes so double-check it and use code with caution. [Learn more](#)

Gemini



Python palindrome checking function:

```
def is_palindrome(s):  
    s = s.lower()  
    return s == s[::-1]
```

- `def is_palindrome(s):`
 - This line defines a Python function named `is_palindrome` that takes one argument, `s`, which is expected to be a string.
- `s = s.lower()`

Gemini



lowercase, we ensure that 'Racecar' and 'racecar' are treated the same.

- `return s == s[::-1]`
 - This is the core of the palindrome check.
 - `s[::-1]` creates a reversed version of the string `s` (which is now all lowercase).

TASK 4

Code Explanation Using Gemini

Prompt Used:

Explain the following Python palindrome checking function line by line:

```
def is_palindrome(s):
```

```
    s = s.lower()
```

```
    return s == s[::-1]
```

The explanation helped me understand slicing operations and how reversing of a string works. Gemini explanation was clear and it was easy to follow.

CONCLUSION –

In this lab, I explored AI – ASSISTED CODING using Google Gemini in Colab. Gemini was helpful for generating beginner-friendly code with detailed explanations. While Co-Pilot provided optimised solutions. Cursor AI helped in refactoring the same. This lab helped me understand how different AI tools assist programmers in different levels.